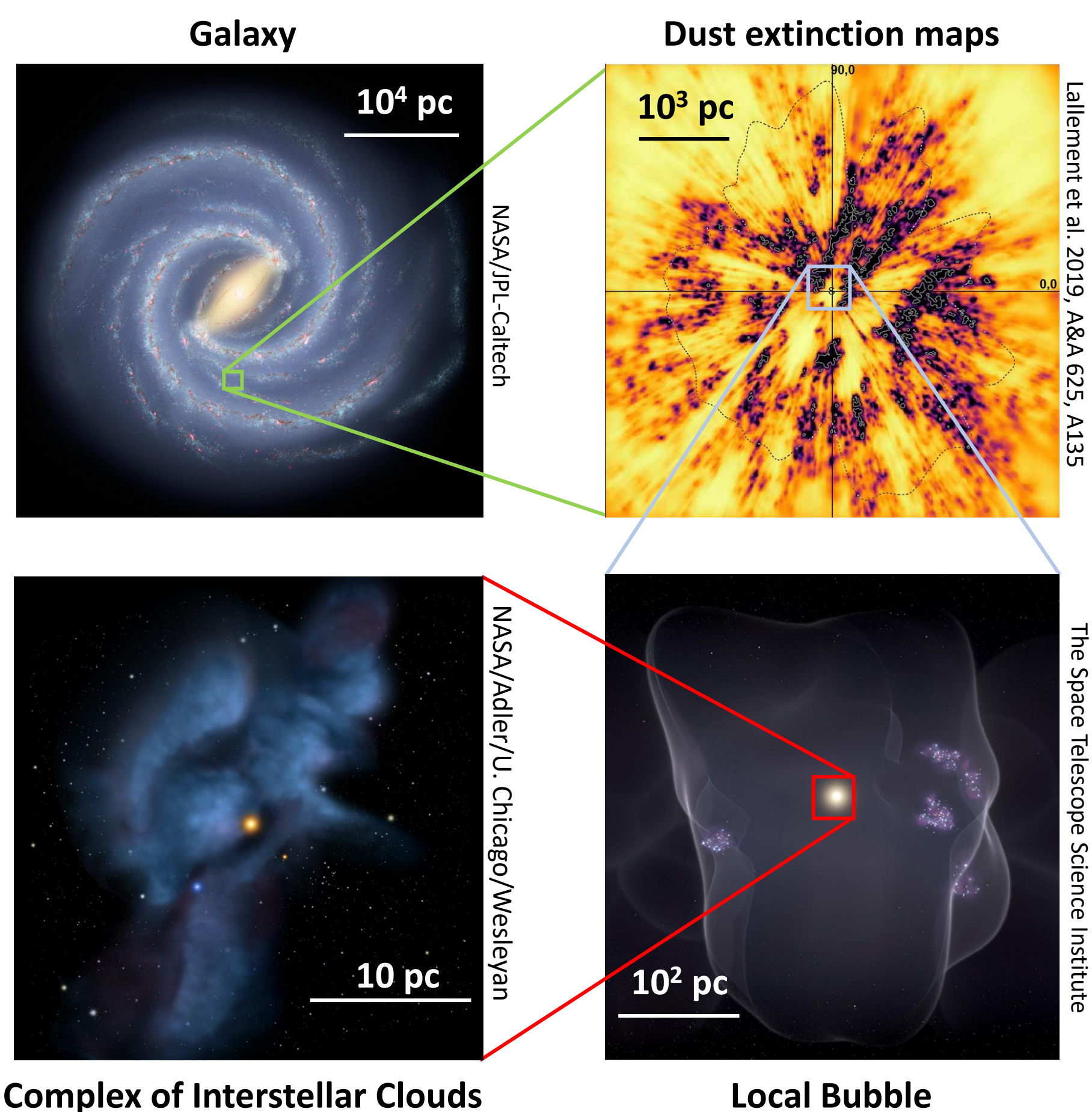
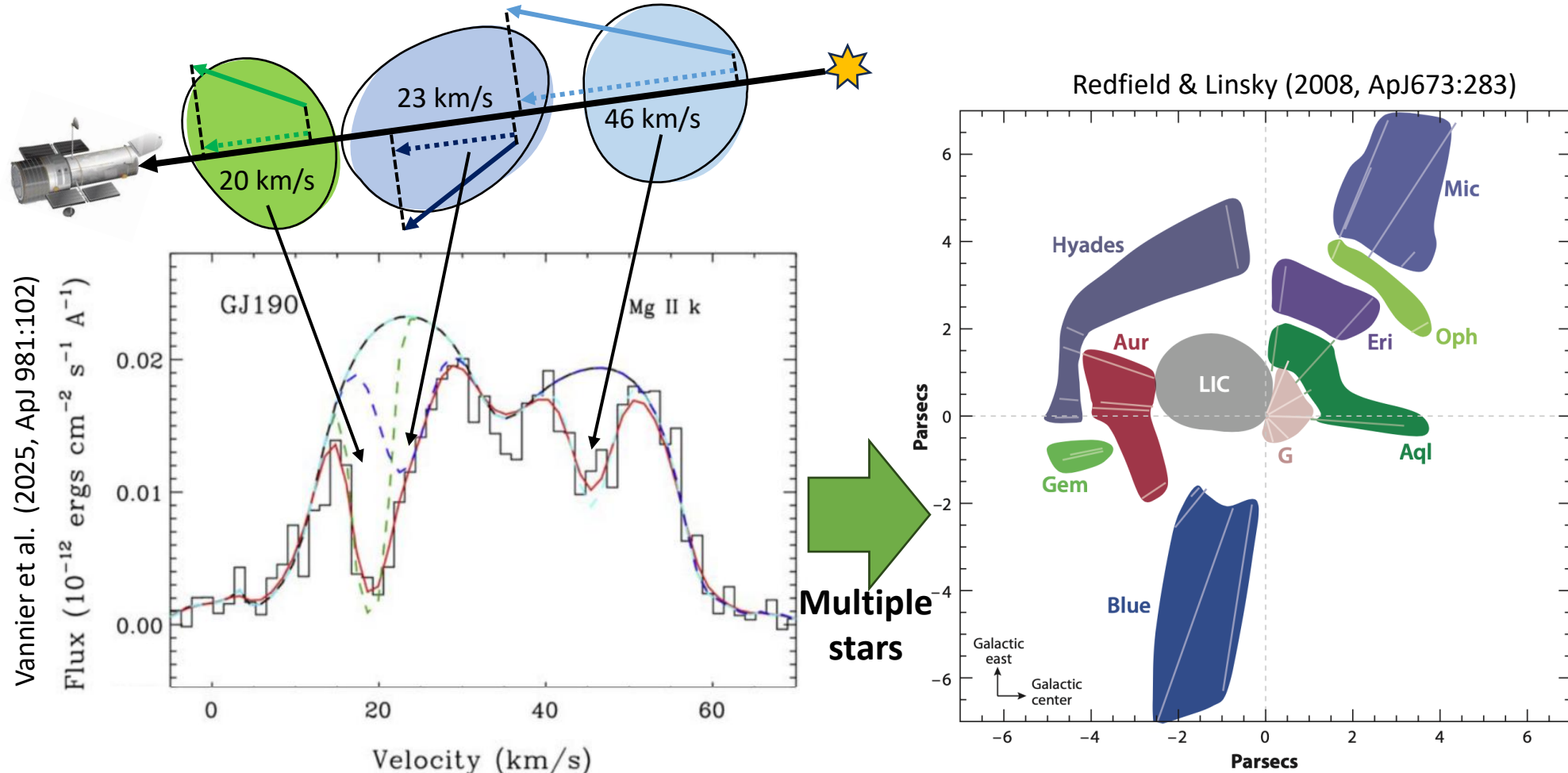




### 1. ASTROPHYSICAL PERSPECTIVE



The smallest structure resolved using astrophysical observations is the complex of 15 interstellar clouds (CLIC) within 15 pc of the Sun.



The Hubble Space Telescope observes absorption lines from the interstellar medium in the spectra of nearby stars. The Sun is located very close to the boundary between the two nearest clouds, the Local Interstellar Cloud (LIC) and the Galactic Cloud (GC), but HST observations cannot resolve the structure of this boundary region.

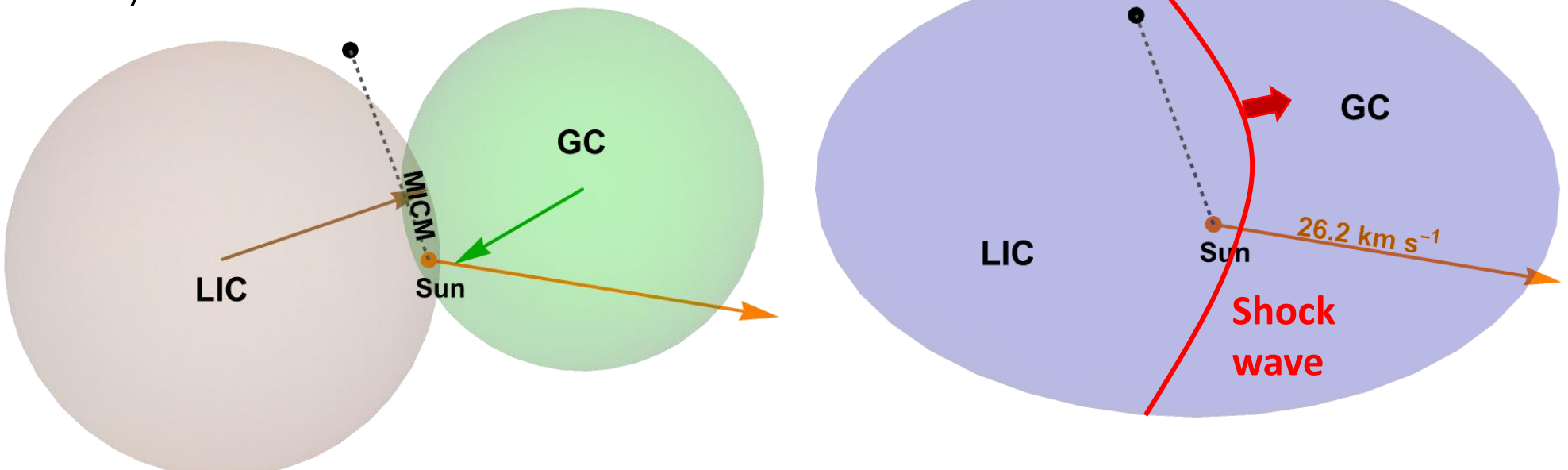
### 4. HYPOTHESES OF LIC-GC INTERACTION

#### Hypothesis I

The LIC and GC velocities suggest that the clouds should collide. It is possible that the Sun is already located in the collision zone (mixing interstellar cloud medium - MICM).

#### Hypothesis II

A shock wave may cross the surrounding interstellar cloud, and the LIC and GC are downstream and upstream plasma, respectively.

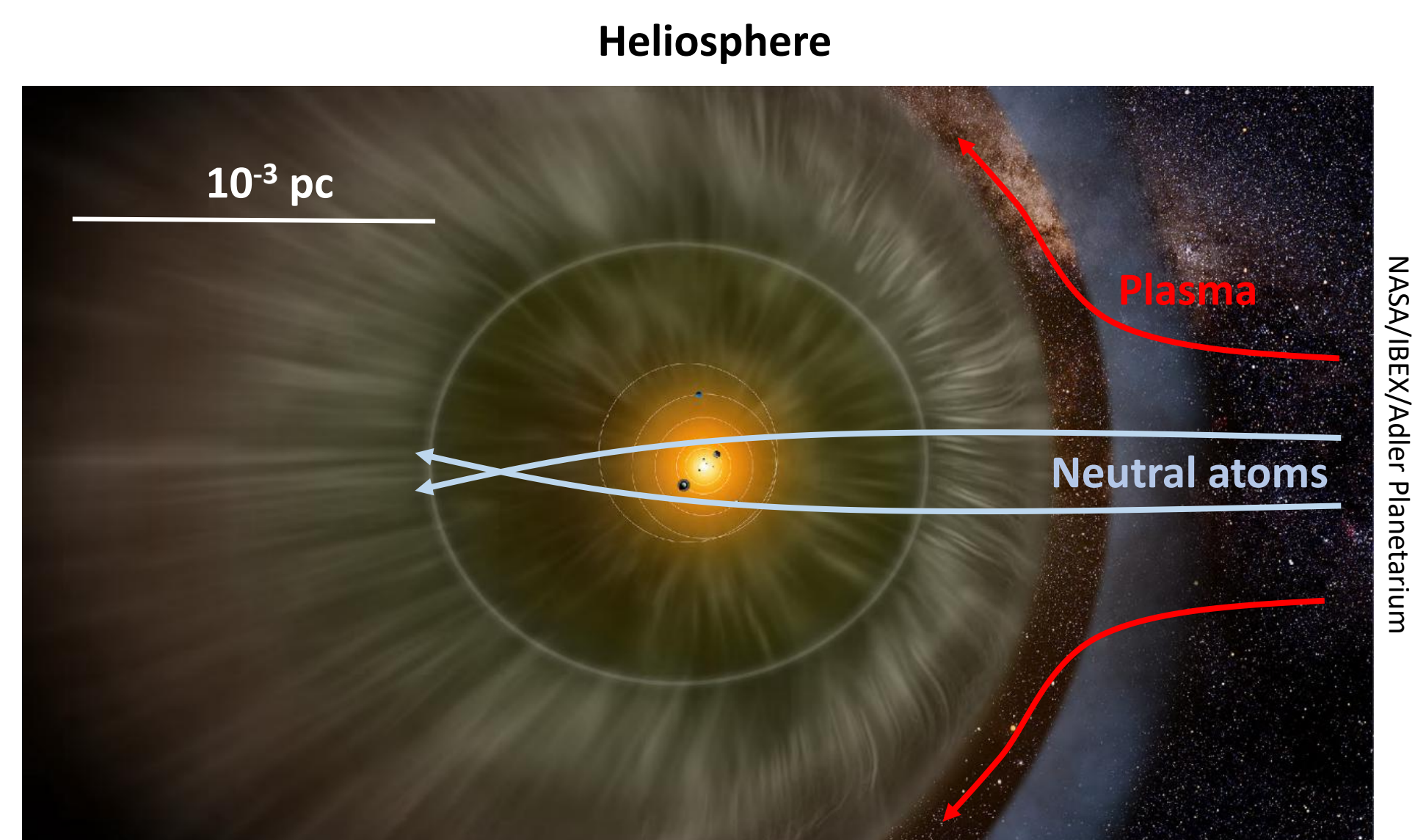


The plasma flow in the mixing zone should be intermediate due to momentum equilibration.

The plasma around the heliosphere may not be equilibrated with the neutrals, especially helium atoms that have a long mean free path between collisions

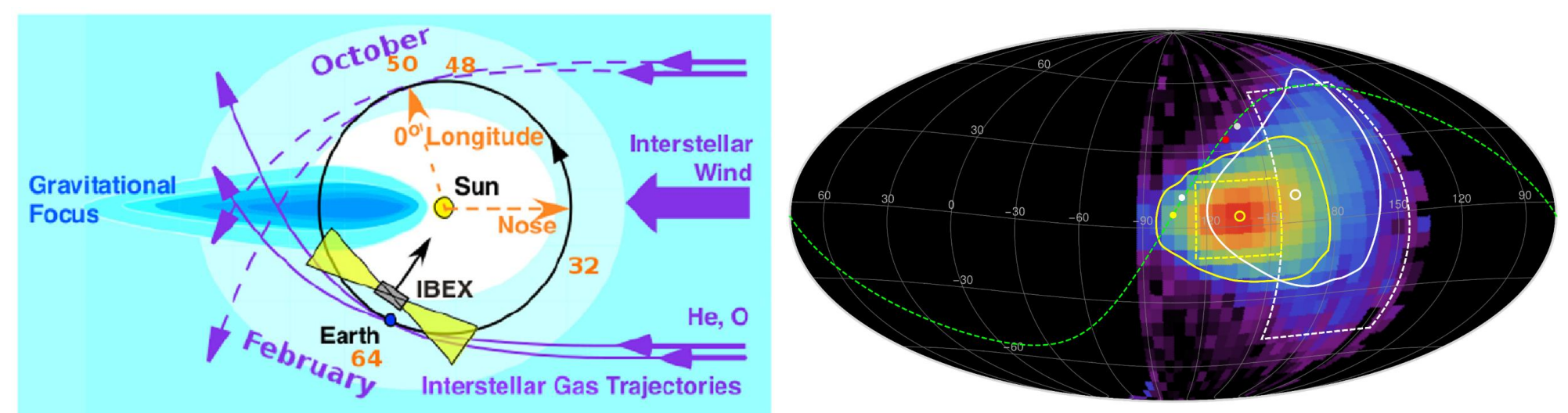
Acknowledgement. This work is funded by the National Science Centre, Poland (NCN), grant 2025/58/E/ST9/00205.

### 2. SPACE PHYSICS PERSPECTIVE



The heliosphere shields the Solar System from most charged particles (except for energetic Cosmic Rays). However, neutral particles, including interstellar neutral (ISN) atoms, enter the heliosphere. Helium atoms are relatively little affected by the heliosphere, and their observations are used to study the local interstellar medium. The IBEX and IMAP missions are equipped with instruments sampling ISN atoms that survive their travel from the LISM to 1 au.

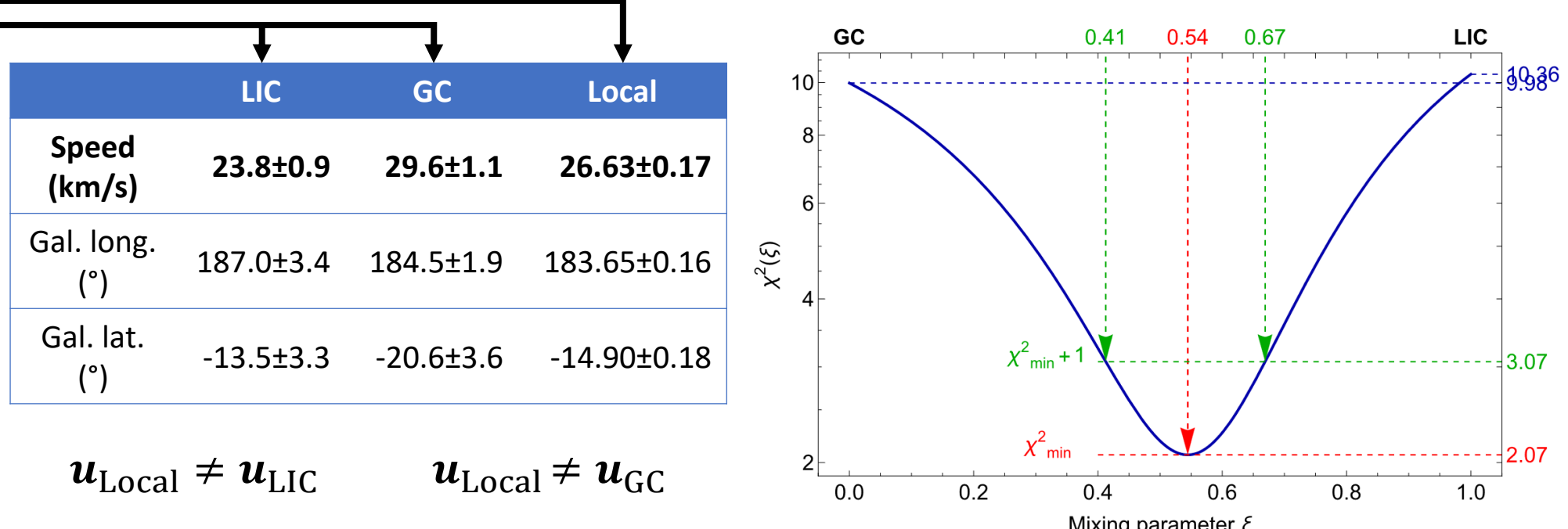
Significant advancements in our understanding of the filtration processes operating in and around the heliosphere have allowed us to limit the systematic bias.



IBEX detects atoms only in the plane perpendicular to the direction to the Sun, resulting in a so-called correlation between flow direction longitude and speed. The recently launched IMAP mission removes this limitation, as its detector sits on a pivot platform.

### 3. COMBINING OBSERVATIONS

Absorption spectra allow us to determine the flow speed in individual clouds, while ISN gas measurements provide the local flow velocity near the Sun. Comparing these values should allow us to determine which cloud the Sun is embedded in.



While neither the LIC nor the GC flow is consistent with the local flow, there is a very good agreement with the linear combination of both velocity vectors:

$$\mathbf{u}_{\text{Local}} \approx \xi \mathbf{u}_{\text{LIC}} + (1 - \xi) \mathbf{u}_{\text{GC}} \quad \xi = 0.54 \pm 0.13.$$

### 5. BRIDGING THE GAP BETWEEN SPACE PHYSICS AND ASTROPHYSICS

Detailed analysis of interstellar neutral gas may help fill a gap in our understanding of our cosmic neighborhood at subparsec scales inaccessible to telescopic observations.

